50Days-coding-challenge

Day-21[SQL]

Insurance Claims Analysis

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A real-world SQL coding challenge on subqueries using an insurance dataset

```
CREATE TABLE
claims (
claim_id INT,
policy_holder
VARCHAR(50),
policy_type
VARCHAR(50),
claim_amount
INT, claim_date
DATE, status
VARCHAR(20)
);
```

```
INSERT INTO claims VALUES
(1, 'Ravi', 'Health', 50000, '2024-01-10', 'Approved'),
(2, 'Anita', 'Vehicle', 20000, '2024-01-12', 'Rejected'),
(3, 'Ravi', 'Health', 30000, '2024-01-20', 'Approved'),
(4, 'Suresh', 'Life', 100000, '2024-01-25', 'Approved'),
(5, 'Anita', 'Vehicle', 15000, '2024-02-01', 'Approved'),
(6, 'Kiran', 'Health', 40000, '2024-02-10', 'Rejected'),
(7, 'Ravi', 'Health', 70000, '2024-02-15', 'Approved'),
(8, 'Suresh', 'Life', 50000, '2024-02-20', 'Rejected');
```

Task 1:

Find claims where **claim amount is greater than average claim amount**

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Query | Query History

```
(8, 'Suresh', 'Life', 50000, '2024-02-20', 'Rejected');

select*from claims;

select*
from claims
where claim_amount > (select avg(claim_amount) as avg_claim_amount from claims);
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	claim_id integer	policy_holder character varying (50)	policy_type character varying (50)	claim_amount integer	claim_date date	status character varying (20)
1	1	Ravi	Health	50000	2024-01-10	Approved
2	4	Suresh	Life	100000	2024-01-25	Approved
3	7	Ravi	Health	70000	2024-02-15	Approved
4	8	Suresh	Life	50000	2024-02-20	Rejected

Total rows: 4 | Query complete 00:00:00.137 | CRLF | Ln 15, Col 1

Task 2:

Find policy holders who have at least one **approved claim**

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Query | Query History

```
SELECT DISTINCT policy_holder
FROM claims
WHERE policy_holder IN (
    SELECT policy_holder
    FROM claims
    WHERE status = 'Approved'
);
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	policy_holder character varying (50)
1	Anita
2	Ravi
3	Suresh

Successfully run. Total query runtime: 152 msec. 3 rows affected.

Total rows: 3 | Query complete 00:00:00.152 | CRLF | Ln 26, Col 3

Task 3:

Find claims where **claim amount is greater than that policy holder's average claim**

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No limit

Query Query History

Save File Ctrl S

```
SELECT *
FROM claims c1
WHERE claim_amount > (
    SELECT AVG(c2.claim_amount)
    FROM claims c2
    WHERE c2.policy_holder = c1.policy_holder
);
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	claim_id integer	policy_holder character varying (50)	policy_type character varying (50)	claim_amount integer	claim_date date	status character varying (20)
1	2	Anita	Vehicle	20000	2024-01-12	Rejected
2	4	Suresh	Life	100000	2024-01-25	Approved
3	7	Ravi	Health	70000	2024-02-15	Approved

Total rows: 3 Query complete 00:00:00.122 CRLF Ln 28, Col 1

Task 4:

Display each claim along with **overall average claim amount**

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No limit

Query Query History

```
SELECT *,
       (SELECT AVG(claim_amount) FROM claims) AS overall_avg_claim
FROM claims;
```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

	claim_id integer	policy_holder character varying (50)	policy_type character varying (50)	claim_amount integer	claim_date date	status character varying (20)	overall_avg_claim numeric
1	1	Ravi	Health	50000	2024-01-10	Approved	46875.000000000000
2	2	Anita	Vehicle	20000	2024-01-12	Rejected	46875.000000000000
3	3	Ravi	Health	30000	2024-01-20	Approved	46875.000000000000
4	4	Suresh	Life	100000	2024-01-25	Approved	46875.000000000000
5	5	Anita	Vehicle	15000	2024-02-01	Approved	46875.000000000000
6	6	Kiran	Health	40000	2024-02-10	Rejected	46875.000000000000
7	7	Ravi	Health	70000	2024-02-15	Approved	46875.000000000000
8	8	Suresh	Life	50000	2024-02-20	Rejected	46875.000000000000

Total rows: 8 Query complete 00:00:00.114 CRLF Ln 36, Col 1